QuickStudy: An Arabic-First Mobile Application for Reading and Studying Lecture Material

Requirements and Design Report · October 2026

Abstract. This report presents QuickStudy, an Android application that combines a PDF reader with study tools for Arabic-speaking university students. It defines the problem and the proposed solution, specifies the functional and non-functional requirements, compares the application with two existing products (Quizlet and Knowt), and presents low-fidelity wireframes of the main screens.

1. Introduction

1.1 Problem Statement

University students commonly study from PDF lecture slides and scanned textbooks. Reading, annotating and self-testing are usually done in separate applications. Existing study platforms are designed primarily for English and offer limited support for Arabic text, right-to-left (RTL) layouts and scanned Arabic documents.

1.2 Proposed Solution

QuickStudy integrates the full study cycle in one application: *upload, read and annotate, organise by subject, and review*. The reader supports notes, highlights and on-device optical character recognition (OCR) for scanned pages. Course files are organised into subjects and chapters, each linked to flashcards and quizzes. A later phase will generate study material automatically using artificial intelligence (AI) through a secure proxy server.

1.3 Scope and Target Users

The system targets Arabic-speaking students using Android phones. It operates offline and stores all files locally. Collaborative features and community-shared content are out of scope.

2. Functional Requirements

ID	Requirement
FR-1	The system shall allow the user to upload PDF and image files and store them locally.
FR-2	The system shall display library books on shelves with cover previews and reading progress.
FR-3	The system shall open a document at the last page read, in vertical or horizontal reading mode, with zoom.
FR-4	The system shall allow the user to add typed notes, highlights and bookmarks to a document.
FR-5	The system shall apply OCR to scanned pages so that their text can be selected.
FR-6	The system shall provide reading settings: brightness, margins, paper colour and night mode.
FR-7	The system shall allow the user to create, edit and delete subjects.

FR-8	The system shall allow the user to add, rename, reorder and delete chapters within a subject.
FR-9	The system shall keep study files separate from library files.
FR-10	The system shall provide flashcard and quiz modes for each chapter.
FR-11	The system shall support Arabic and English interfaces and light and dark themes.
FR-12	The system shall generate summaries, flashcards and quizzes from a chapter using AI *(future phase)*.

3. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Performance	A page shall render within one second on a mid-range device, and heavy processing shall run in the background.
NFR-2	Availability	All features except AI generation shall work offline.
NFR-3	Privacy and Security	User files shall remain on the device, and no secret API key shall be embedded in the application.
NFR-4	Localization	The interface shall fully support Arabic, including RTL layout and Arabic plural forms.
NFR-5	Usability and Accessibility	Touch targets shall be at least 44 px, and every screen shall define loading, empty and error states.
NFR-6	Reliability	Database upgrades shall preserve existing user data.
NFR-7	Maintainability	The code shall follow the Model–View–ViewModel (MVVM) architecture with one-way data flow.

4. Comparative Analysis

Quizlet is a widely used flashcard platform with a large library of shared study sets and an AI tutor available to paid subscribers. It does not include a document reader, and its Arabic support is limited.

Knowt generates notes, flashcards and practice tests from uploaded files using AI. Files are processed in the cloud, and AI usage is limited on the free plan.

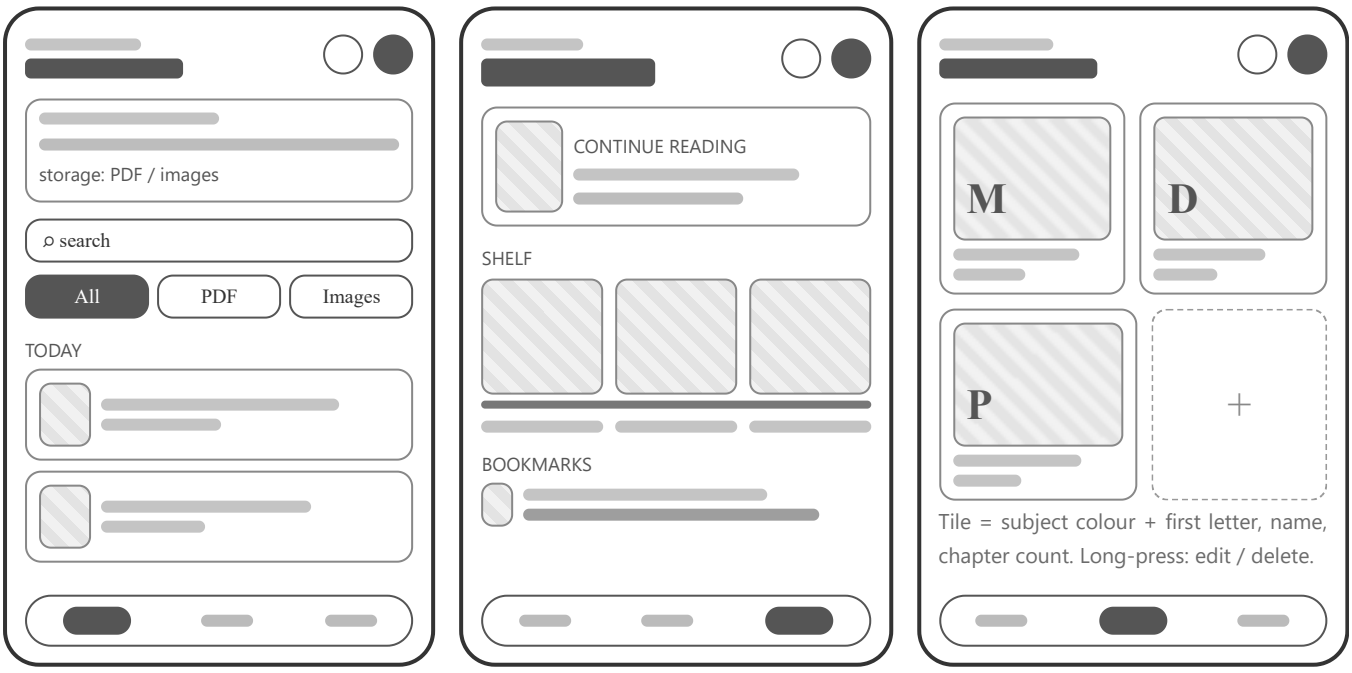
Criterion	QuickStudy	Quizlet	Knowt
Integrated PDF reader with annotation	Yes	No	Limited
OCR for scanned pages	On device	No	Cloud
Full Arabic and RTL support	Yes	Partial	Partial
AI generation from user files	Planned	Paid tier	Yes
Offline use and local storage	Yes	Partial	Partial
Shared community content	No	Yes	Yes

Table 1. Feature comparison of QuickStudy and two competing applications.

QuickStudy is distinguished by its integrated reader, its Arabic support and its local storage of user files. AI-based generation of study material is required to match the main capability of both competitors.

5. Low-Fidelity Wireframes

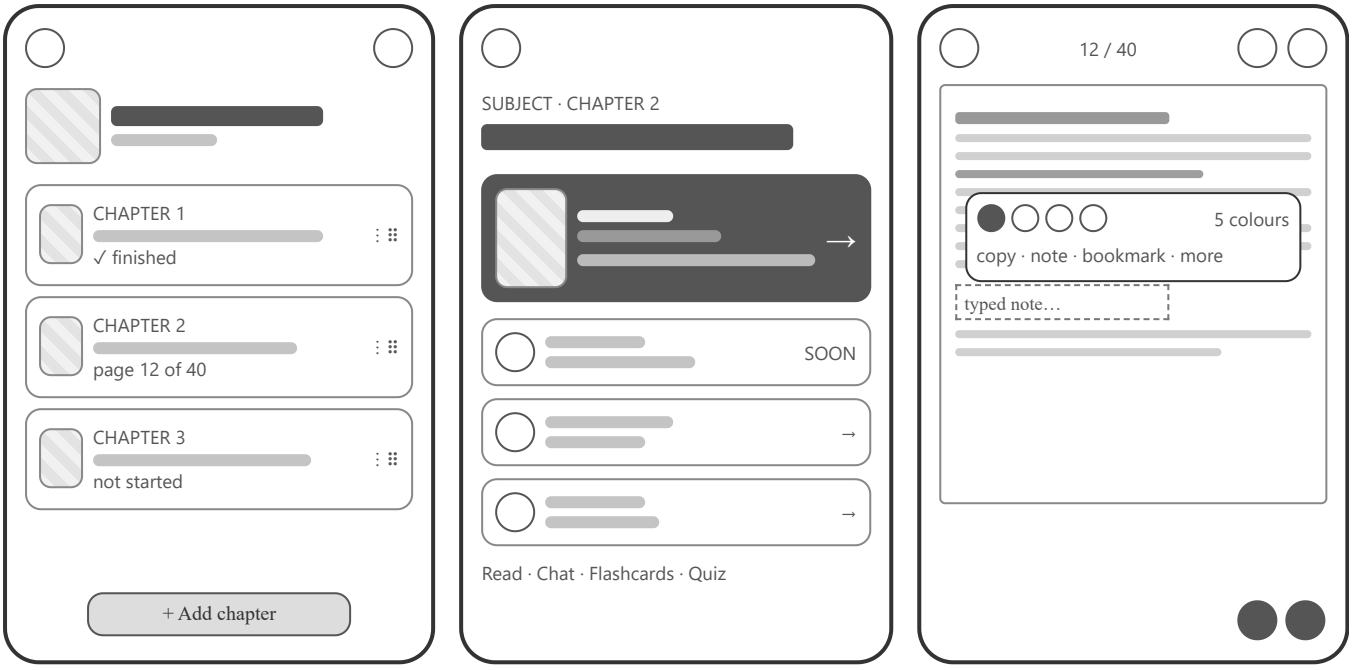
Figure 1 presents grayscale wireframes of the six main screens. They show layout and hierarchy only.



Home (files)
Upload, storage, search, filter, swipe to delete

Library (shelves)
Covers + progress, continue reading, bookmarks

Study (subjects)
Subjects work like folders



Subject (chapters)
Same-height rows, drag to reorder, status line

Chapter
One featured action (Read), then study modes

Reader
Highlight menu, typed note, zoom buttons

Figure 1. Low-fidelity wireframes of the main screens (LTR layout; the Arabic layout is mirrored).

6. Conclusion

QuickStudy addresses the separation between reading and studying, and the limited Arabic support in current study applications. The core reading and organisation features are implemented. Future work will add AI-generated study material through a secure server, and an evaluation with student users.

References

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